

AI Principles and Practice · Lecture 3

Ethics, Bias and Fairness

Week 3 · Phase I — Foundations and Ethics · LOs: LO5, LO3

Lecturer: Jonathan

Where today goes

1. **Foundations** — why fairness is a modelling decision, not a postscript
2. **Where bias lives** — data, algorithm, society, and the joins between them
3. **Fairness, made precise** — parity, odds, impact, calibration
4. **The catch** — why you can't have every fairness at once
5. **The lab** — audit a real model, break it open, cost the fix

The through-line: one credit model, followed all the way from base rates to the price of “fixing” it.

What you'll be able to do by the end of today

These map to **LO5** (ethics, fairness, governance) and **LO3** (evaluate and compare models). Note the verbs — none of them is “define”.

- **Locate** where bias enters a pipeline you’re handed, and argue which entry point matters most for a given use.
- **Judge** a model against several fairness criteria at once, and defend the one you’d hold it to.
- **Reason** about why the criteria collide — not as a bug, but as a result you could have predicted.
- **Cost** a mitigation honestly: what it buys, what it spends, and who ends up paying.
- **Compose** an ethical-impact assessment that a sceptical stakeholder would actually accept.

Start with a decision, not a definition

A bank runs a model to approve loans. It never sees anyone's ethnicity — that column was deleted on principle.

Twelve months on, one group is approved at **58%**, the other at **22%**. Same model. Same rules. No protected attribute anywhere in it.

Is that discrimination? A data problem? Just the world showing through? Hold the question — by the end you'll have four defensible answers and a reason they can't all be satisfied.

The uncomfortable part

Nobody in this story did anything obviously wrong. The unfairness is **emergent** — it lives in the interaction of data, objective and world, not in a villain.

Your task today isn't to feel bad about this. It's to **measure** it, **argue** about it, and **decide** what to do.

Section 1

Fairness is a modelling concern

Ethics isn't the final slide

Most courses bolt ethics on at the end — a week of cautionary tales after the “real” work is done. That framing is the problem.

If ethics comes last, it’s decoration. By the time the model ships, every consequential choice — what to predict, on whom, optimised for what — is already frozen.

Fairness is decided upstream, in the same breaths as feature selection and loss design. It’s an engineering variable, not a compliance sign-off.

So we’re doing it in Week 3, next to evaluation — because that’s where the decisions that fix fairness are actually made.

A model is an argument about the world

Every model encodes claims: that these features matter, that this target is worth predicting, that this error is more tolerable than that one.

- **Choosing the target** says what counts as success.
- **Choosing the loss** says which mistakes you'll accept.
- **Choosing the data** says whose past you'll generalise from.

None of those is neutral. Pretending they are is the first ethical failure.

Reframe

Don't ask "is this model biased?" as a yes/no. Ask "what does this model **assume**, and who benefits if the assumption holds?"

"Just maximise accuracy" is already a value choice

Accuracy treats every error the same. Reality never does.

A model that's 95% accurate can be catastrophic if the 5% it fails on are all the same people. Accuracy averages that away — that's the whole danger.

Optimise for the aggregate and you quietly decide that a rare, concentrated harm is an acceptable price for a good headline number.

Interrogate the objective

Before you accept any metric, ask what it's willing to sacrifice — and whether you'd say that out loud to the people it sacrifices.

Three ethical lenses on a single threshold

Move a decision threshold and you can watch each lens light up differently. They rarely agree.

Lens	Asks of the model	Where it bites
Consequentialist	Does it produce better outcomes overall?	Ignores how harm is distributed
Deontological	Does it treat people as ends, by fair rules?	May reject a model that helps the average person
Justice / Rawlsian	Does it improve the position of the worst-off?	Can demand you give up aggregate performance

Pick your lens **before** you tune, not after the numbers come back. Otherwise the metric picks the ethics for you.

Whose fairness are we even talking about?

“Fair” is not one thing. The applicant, the lender, the regulator and the rejected-but-creditworthy borrower all want something different — and some of those wants are mutually exclusive.

- The **applicant** wants an equal shot.
- The **lender** wants precision on repayment.
- The **regulator** wants no disparate impact.
- **Society** wants the historical gap to shrink, not calcify.

Power is the hidden variable

Whoever gets to **define** fairness usually isn't the person the model decides about. Notice who's in the room when the metric is chosen.

From "accurate" to "defensible"

The shift this section asks of you:

Old question

How accurate is it? A single number.
Easy to report, easy to hide behind.

Better question

Could you defend this decision to the person it was made about? Now you need distribution, not just average.

Everything that follows is machinery for answering the second question with evidence instead of vibes.

Section 2

Where bias actually enters

"Bias" is three different words in a trench coat

We use one word for three unrelated things, then talk past each other. Untangle them first.

Sense	Means	Fix is
Statistical	estimator systematically off the true value	more/better data, better model
Social	systematically worse outcomes for a group	not always technical — sometimes structural
Cognitive	human judgement skewed by heuristics	process, review, diverse teams

Why it matters

Reducing **statistical** bias can **increase social** bias — a more accurate model of an unjust world reproduces the injustice more faithfully.

Data bias: the world writes your training set

The data isn't a neutral photograph. It's a record of who got measured, who got counted as a success, and whose story never made it in.

- **Sampling** — who's over- or under-represented.
- **Historical** — past decisions baked in as “ground truth”.
- **Measurement** — the proxy you could record $\neq$ the thing you meant.
- **Labelling** — someone's judgement, with all its slant, frozen as a target.

Diagnose, don't assume

For any dataset you're handed: which of these four is most likely, and how would you **test** for it rather than guess?

The label is usually the real crime scene

We obsess over features and trust the label. Backwards.

You wanted to predict **creditworthiness**.
What you actually have is “**did the bank approve them last time and did they repay**” — a record shaped by who was offered credit at all.

Train on that and you don’t learn who’s creditworthy. You learn **who the old system already favoured** — then launder it as objective.

Proxy label

When the target is a stand-in for what you really care about, audit the gap between them **before** you touch a metric. That gap is where the bias hides.

Algorithmic bias: the objective is an opinion

Even with clean data, the algorithm adds its own slant.

- **Objective functions** reward whatever you told them to — including shortcuts you didn't mean.
- **Aggregation** optimises the average, which is a choice about whose errors are visible.
- **Regularisation** nudges toward the majority pattern, smoothing minorities away.

The optimiser is faithful, not malicious. It gives you exactly what you specified — which is the problem when your specification was a compromise you didn't examine.

Societal bias: you're modelling an unequal world

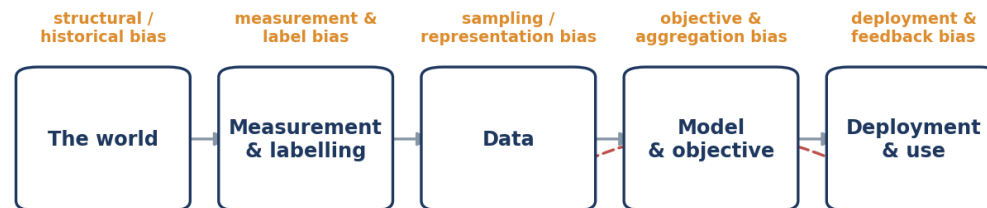
Some gaps in the data are real, in the sense that they're truly there — and they're there because the world that produced them was unequal.

A model can be **perfectly accurate** about that world and still be an instrument for entrenching it. Accuracy and justice are different axes.

The hard question

Should a model reflect the world as it **is**, or nudge toward the world as it **should be**? There's no neutral answer — refusing to choose is itself a choice for the status quo.

Map the whole pipeline before you blame the model



feedback loop — yesterday's decisions become tomorrow's training data

Bias enters at every stage — and the feedback loop means today's decisions become tomorrow's “ground truth”.

When someone says “the algorithm is biased”, make them point at the stage. Nine times in ten it's upstream of the model — and blaming the model lets the real cause off the hook.

Deleting the sensitive column doesn't work

Our credit model **never sees** group membership. “Fairness through unawareness”, supposedly.

Yet from the model’s **score alone**, you can recover which group someone is in **70% of the time** — against a **42%** base rate.

Income, credit history and default counts are **proxies**. The model reconstructs the very attribute you tried to hide.

Evaluate the claim

“We don’t use race/gender” is a comfortable defence and a weak one. Redundant encoding means the attribute is **in there** whether you named it or not. Blindness is not fairness.

Feedback loops: the model edits its own future

Deny loans to a group and you never observe their repayments. Next year's training data now has **even less** signal about them — so the model gets **more** confident and **more** wrong.

The disparity isn't just measured by the system; it's **manufactured** by it. A snapshot audit misses this entirely — you have to reason about the loop over time.

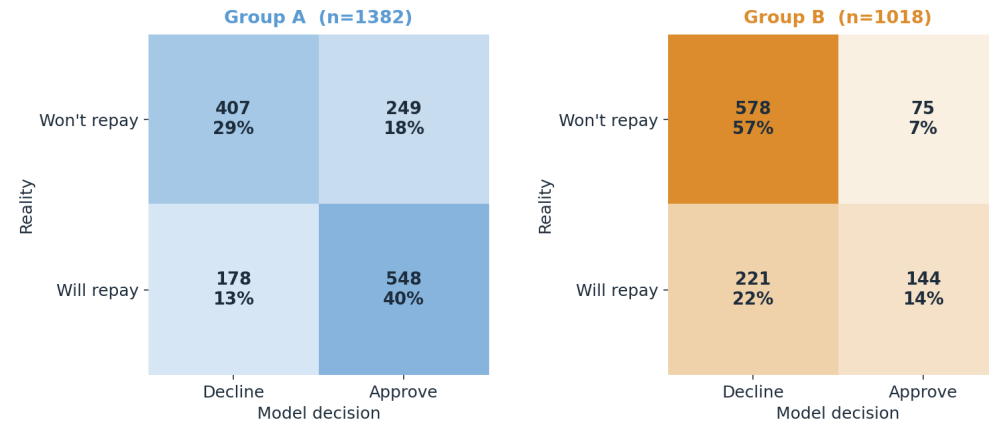
Higher-order prompt

Sketch how a “predictive policing” model could look **more** accurate every year while describing reality **less** well. What is it actually measuring?

Section 3

Fairness, made precise

Week 2's confusion matrix, now split by group



Same model, one matrix per group. Every fairness metric today is just a comparison between these two tables.

This is the whole trick: don't compute one confusion matrix, compute one **per group**, then argue about which cell you're allowed to let differ.

Two families, and they don't get along

Group fairness

- Compares **rates** across groups.
- “Approve A and B at similar rates / with similar error.”
- Easy to measure. Blind to the individual.

Individual fairness

- “Similar people, similar treatment.”
- Intuitive and hard — you must define **similar**, which smuggles values right back in.

The catch you'll meet all lecture

Enforcing group fairness can treat **similar individuals differently** across groups.
Enforcing individual fairness can leave **group** rates wildly apart. Pick your poison deliberately.

Demographic parity — and its blind spot

$$P(\hat{Y} = 1 \mid A) = P(\hat{Y} = 1 \mid B)$$

the chance of approval is the same whichever group you're in — it looks only at the decision, never at who deserved it

Same **approval rate** for every group. Simple, legible, beloved of regulators.

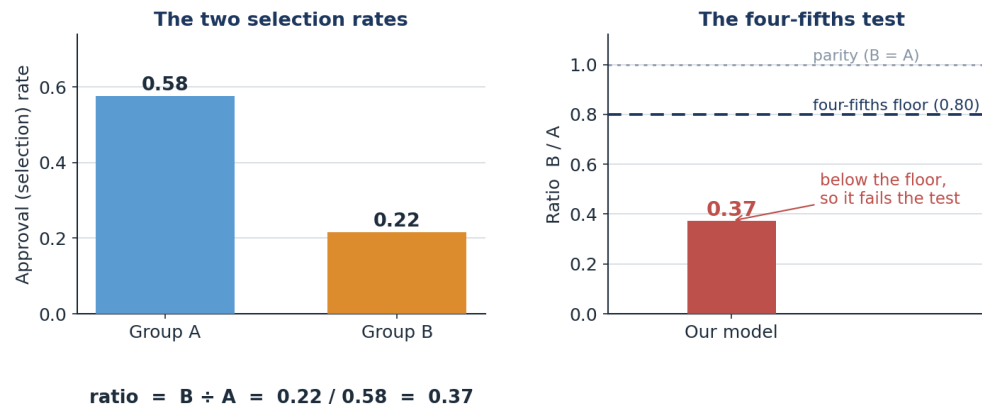
Its blind spot: it ignores **who actually repays**. If the groups genuinely differ in outcome, parity can force you to approve worse-qualified applicants from one side — helping nobody and satisfying a formula.

When it's right

Use it when you believe the label itself is corrupted — when differing base rates reflect **past injustice**, not real difference. That belief is an argument you must make, not assume.

Disparate impact and the four-fifths rule

A legal test, not just a statistical one. If the disadvantaged group's selection rate falls below **0.80** of the advantaged group's, that's **prima facie** disparate impact.



Read it in **ratio** space, not rate space: the rule asks whether $B \div A$ clears **0.80**. Ours is **0.37**, under half the floor. It wouldn't survive a tribunal, let alone an ethics review.

Equalised odds — fairness on the errors

$$\text{TPR}_A = \text{TPR}_B \quad \text{and} \quad \text{FPR}_A = \text{FPR}_B$$

the errors mustn't depend on your group: equal approval among those who truly repay, equal wrong-approval among those who don't

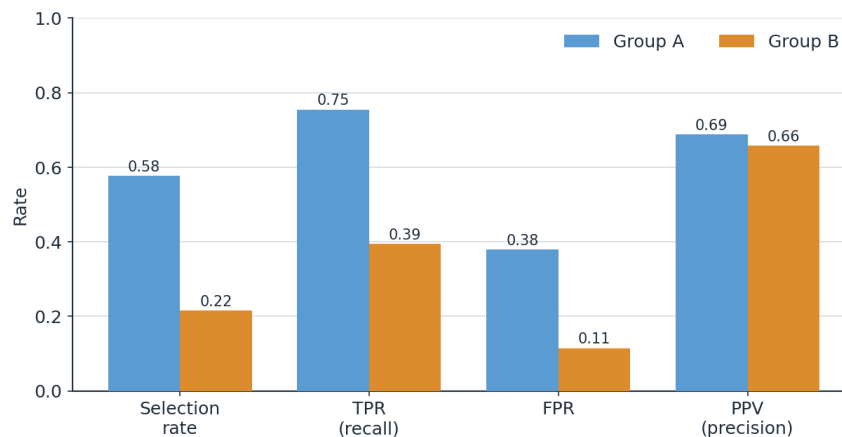
Equal opportunity relaxes this to the TPR alone — the “don't miss the deserving” half.

Our model: TPR **75%** vs **39%**. A qualified Group-B applicant is **nearly half** as likely to be approved as an equally qualified Group-A one.

Why prefer this to parity

Equalised odds **conditions on the truth**. It doesn't demand equal approvals — only that being right or wrong doesn't depend on your group. Often the more defensible ask.

The gaps, side by side



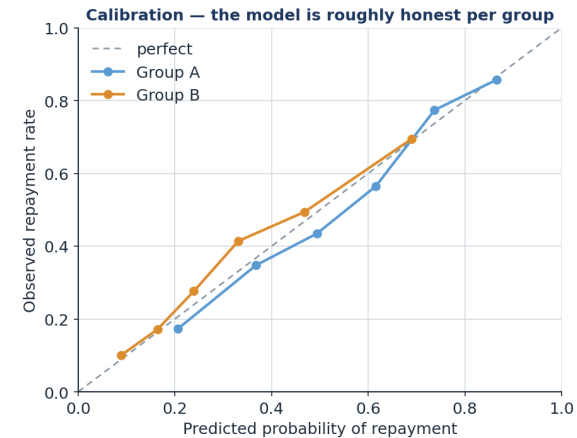
*One model, four metrics, two groups. The bars **are** the ethical argument — now you have to decide which gap is unacceptable.*

Notice precision (PPV) is almost level — **69%** vs **66%** — while selection and TPR are miles apart. That near-match is about to become the trap.

Calibration: the model is "honest" per group

A score is **calibrated** if “0.7” means a 70% repayment rate — for **both** groups. Ours roughly is: within each group, the scores mean what they say.

So the model is simultaneously **well-calibrated** and **wildly unequal** on selection and TPR. Both statements are true. Sit with that.



Same model, four verdicts

Criterion	Our model	Verdict
Demographic parity	58% vs 22%	Fails badly
Disparate impact (0.80)	ratio 0.37	Fails — legally exposed
Equal opportunity (TPR)	75% vs 39%	Fails
Calibration / precision	69% vs 66%	Passes

The evaluative move

The **same** model is fair or unfair depending on the metric. “Is it fair?” is unanswerable. “Fair by which criterion, and can you justify that choice?” is the real question — and it’s the one your assessment must answer.

Section 4

Why you can't have it all

The impossibility result, in plain words

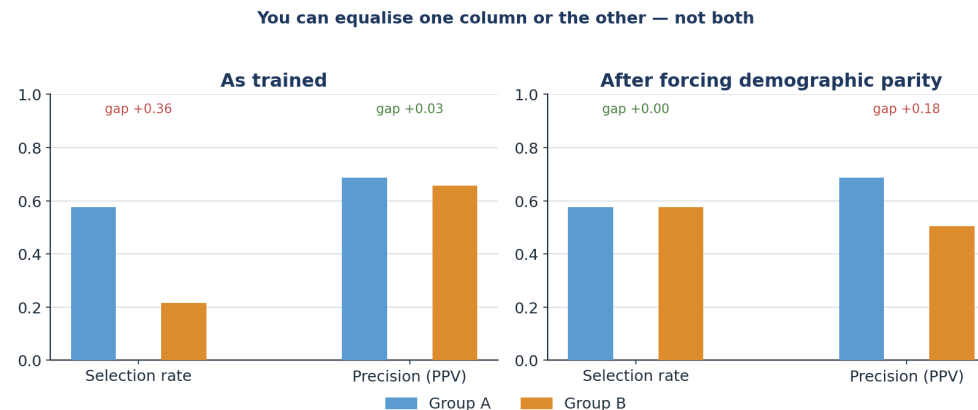
When two groups have **different base rates**, you **cannot** have all of: equal calibration, equal false-positive rates, and equal false-negative rates. Not “it’s hard” — mathematically ruled out.

This isn’t a limitation of your model or your effort. It’s a theorem (Kleinberg, Mullainathan, Raghavan 2016; Chouldechova 2017).

So what breaks

Any tool promising to make a model “fair” full stop is selling you a **choice disguised as a solution**. It picks which fairness to satisfy and quietly drops the rest.

Watch the trade happen



Left: as trained — selection differs, precision roughly matches. Right: force equal selection and B’s precision collapses 66% → 50%. You moved the unfairness; you didn’t remove it.

This is the whole lecture in one picture. Every “fix” is a **transfer** of unfairness from one metric to another. The honest question is which transfer you can defend.

What algorithmic justice can — and can't — do

It can

Make the trade-offs **visible** and **quantified**; force an explicit, auditable choice; catch the disparities a single accuracy number hides.

It can't

Invent a fairness that satisfies everyone; repair injustice that lives in the data, not the model; substitute for the political decision about which harm is worse.

The metric doesn't resolve the ethics. It **sharpens** it — turning a vague unease into a specific, arguable decision. That's the most it can do, and it's a lot.

Fairness is a political choice wearing a metric

Choosing demographic parity over equalised odds isn't a technical preference. It's a stance on whether unequal base rates are **real** or **the residue of injustice**.

That stance belongs to stakeholders, ethicists, the affected public — not to whoever happened to write the training script.

Your professional line

Your job is to make the choice **legible and honest**, surface its costs, and refuse to let a default metric make a moral decision by accident.

Section 5

The lab — audit, break, cost

Today's lab, end to end

You'll take the Week 3 credit model and put it through a real audit, end to end.

1. **Measure** group metrics with Fairlearn or AIF360 — selection, TPR/FPR, DI, calibration.
2. **Attempt** one mitigation and **quantify its cost** — in accuracy, and in the fairness metrics it **doesn't** fix.
3. **Write** a short ethical-impact assessment: what you found, what you changed, what you'd still refuse to ship.

The deliverable isn't a "fair model" — there's no such thing. It's a defended decision with the receipts attached.

Fairlearn or AIF360 — pick your tool

	Fairlearn (Microsoft)	AIF360 (IBM)
Feel	lightweight, sklearn-native	broad, research-grade
Metrics	MetricFrame — clean per-group tables	70+ metrics, exhaustive
Mitigation	pre / in / post, few strong methods	many algorithms to compare
Start here if	you want an answer this afternoon	you want to survey the whole space

Either is fine for the audit. Fairlearn gets you to a defensible result faster; reach for AIF360 when you want to **compare mitigations**, not just apply one.

Measuring is five lines, not a library

audit.py

python

```
from fairlearn.metrics import (
    MetricFrame, selection_rate)
from sklearn.metrics import (
    recall_score, precision_score)

mf = MetricFrame(
    metrics={"selection": selection_rate,
            "tpr": recall_score,
            "ppv": precision_score},
    y_true=y_test, y_pred=y_hat,
    sensitive_features=group_test)

print(mf.by_group)
print(mf.difference())
```

mf.by_group / mf.difference()

	selection	tpr	ppv
A	0.577	0.755	0.688
B	0.215	0.395	0.658

difference	
selection	0.362
tpr	0.360
ppv	0.030

The maths isn't the hard part. The skill is **reading** that table: a 0.36 selection gap beside a 0.03 precision gap is the whole ethical argument in three rows.

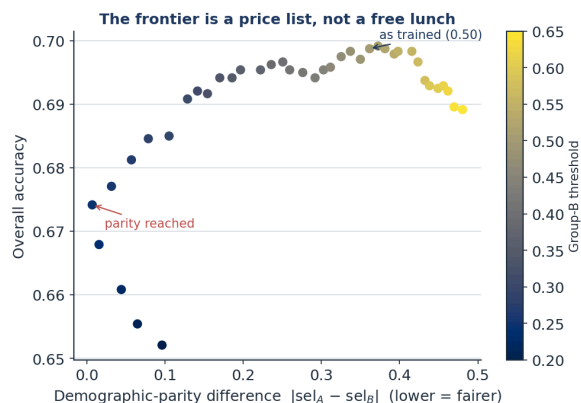
Three places to intervene

Stage	You change	Trade-off
Pre -processing	the data — reweight, relabel, resample	touches truth; can feel like cooking the books
In -processing	the training — fairness as a constraint	most principled, least portable
Post -processing	the thresholds — per-group cutoffs	easiest to bolt on; explicitly unequal treatment

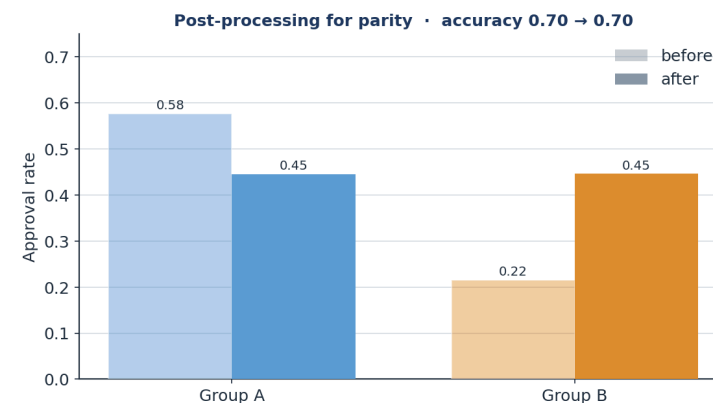
Design decision

None is “the right one”. Post-processing is quickest but means literally applying **different thresholds by group** — defensible, but you’d better be ready to say so publicly.

Cost the fix — and ask who pays



Sweeping toward parity: the frontier is a price list.



Post-processing for parity — approvals rebalance.

The subtle finding

Aggregate accuracy barely moved. But Group A's approvals fell **58% → 45%**. The cost wasn't “accuracy” — it was **redistribution**. Always ask **who pays**, not just **how much**.

The ethical-impact assessment

A page, not a treatise. Structure it so a sceptic can't wave it away:

- **Context** — what the model decides, about whom, and what's at stake for them.
- **Findings** — the group metrics, stated plainly, worst gap first.
- **Choice** — which fairness criterion you're holding it to, and **why that one**.
- **Cost** — what your mitigation spent, and who bore it.
- **Residual risk** — what you did **not** fix, and what would make you refuse to ship.

What separates a strong audit from a box-tick

Box-tick

- Reports one fairness number.
- Declares the model “fair”.
- Hides the trade-off.
- Treats mitigation as a fix.

Strong

- Shows several metrics **disagreeing**.
- Names the criterion and defends it.
- Quantifies what the fix **costs**.
- Says what it would **refuse** to deploy.

The grade — and the professional respect — is in the **defence**, not the model. Show me a decision you can stand behind.

What you're handing in this week

In the lab

- Group-metric audit of the Week 3 credit model.
- One mitigation attempt, with its cost quantified.
- A one-page ethical-impact assessment.

Carries into the project

- The audit becomes a **section** of your final report — not an appendix.
- Your project's fairness choice, argued and costed, is assessed.

Keep the audit code you write today. The assignment runs this same workflow on its assigned dataset — where the fairness choice, and its defence, are yours to make.

Five things worth leaving with

1. **Fairness is designed in, not inspected out.** The choices that decide it happen at target, loss and data — upstream of any audit.
2. **Blindness isn't fairness.** Delete the sensitive column and proxies rebuild it — 70% recoverable here from the score alone.
3. **“Fair” is meaningless without a criterion.** One model failed three tests and passed a fourth. Always ask “fair **by which** definition?”.
4. **The trade-offs are a theorem, not a bug.** Different base rates make the criteria mathematically incompatible. Fixing one moves the unfairness; it doesn't delete it.
5. **Cost the fix, and name who pays.** Mitigation is redistribution. Your job is to make that honest, defensible and out loud.

Where to read next

Papers

- Kleinberg, Mullainathan & Raghavan (2016), **Inherent Trade-Offs in the Fair Determination of Risk Scores** — the impossibility result itself.
- Chouldechova (2017), **Fair Prediction with Disparate Impact** — the same collision, framed for practitioners.
- Hardt, Price & Srebro (2016), **Equality of Opportunity in Supervised Learning** — where equalised odds comes from.
- Barocas & Selbst (2016), **Big Data's Disparate Impact** — the legal-technical bridge.
- Buolamwini & Gebru (2018), **Gender Shades** — audit that changed the industry.

Books & tools

- Barocas, Hardt & Narayanan, **Fairness and Machine Learning** — free at fairmlbook.org; the standard text.
- O'Neil, **Weapons of Math Destruction** — the accessible case for why this matters.
- Fairlearn user guide (fairlearn.org) — start with **MetricFrame**.
- AIF360 (aif360.res.ibm.com) — for surveying mitigations.

Read Kleinberg **et al.** before next week. It's short, and it makes the impossibility result feel inevitable rather than surprising.